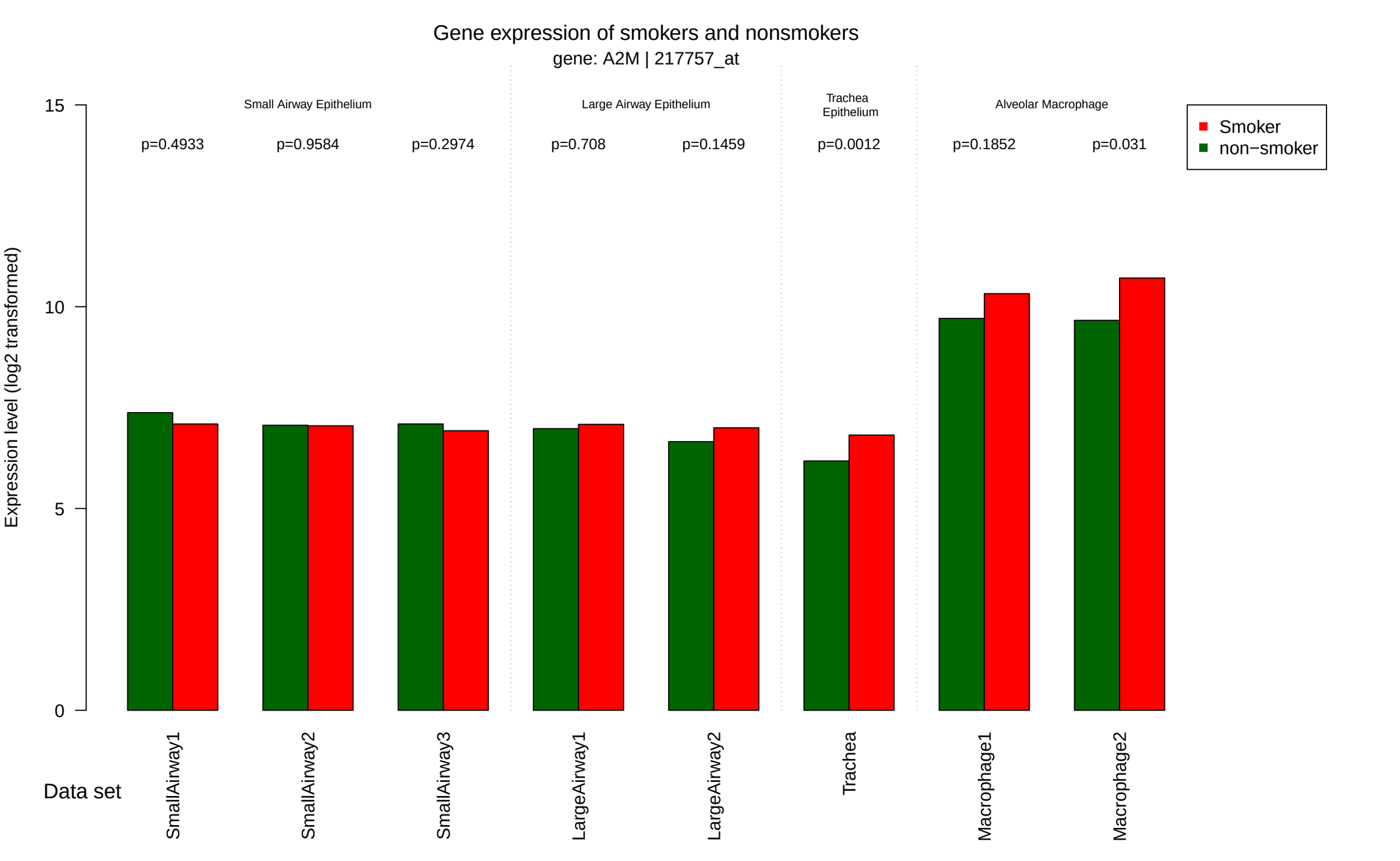
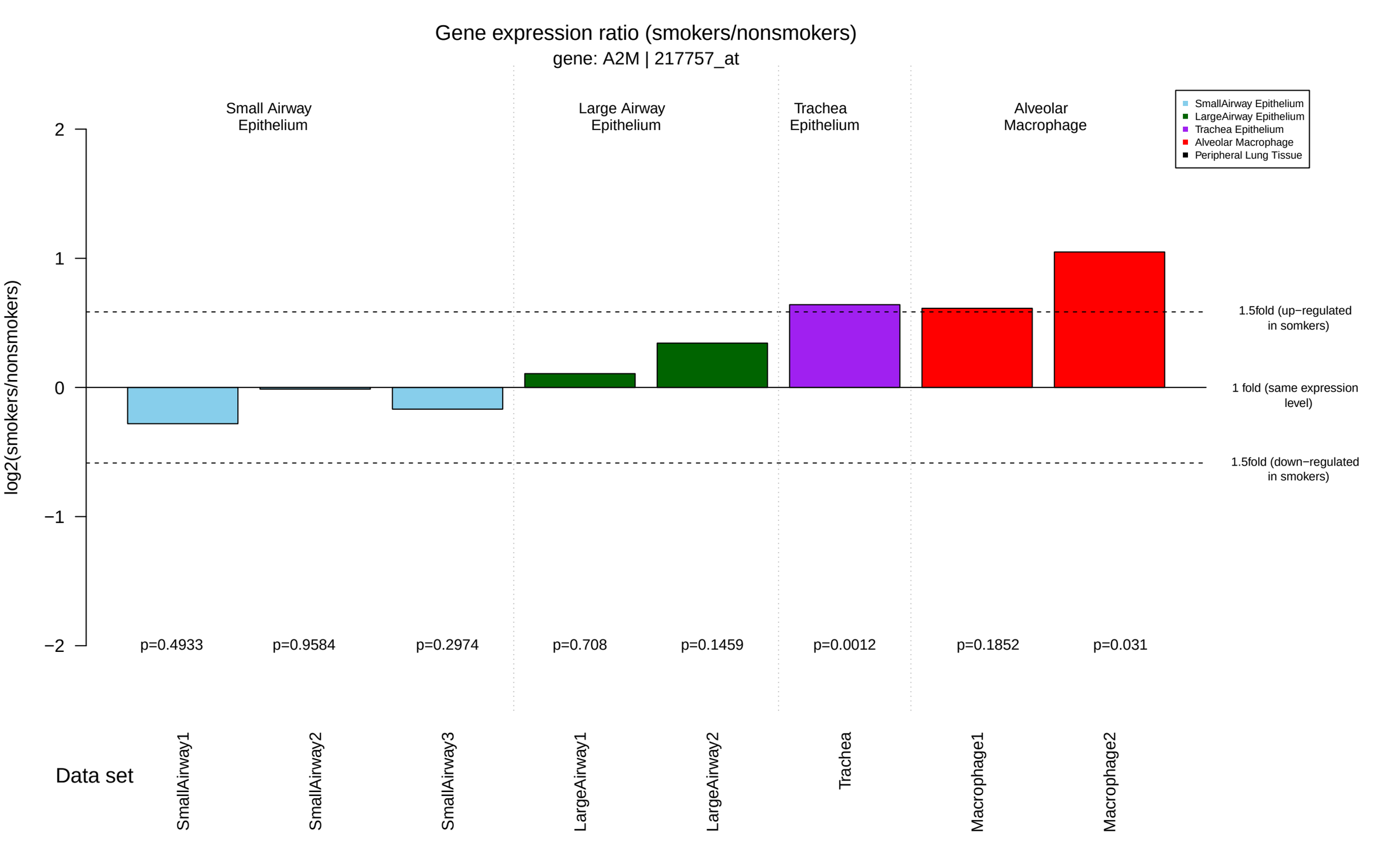
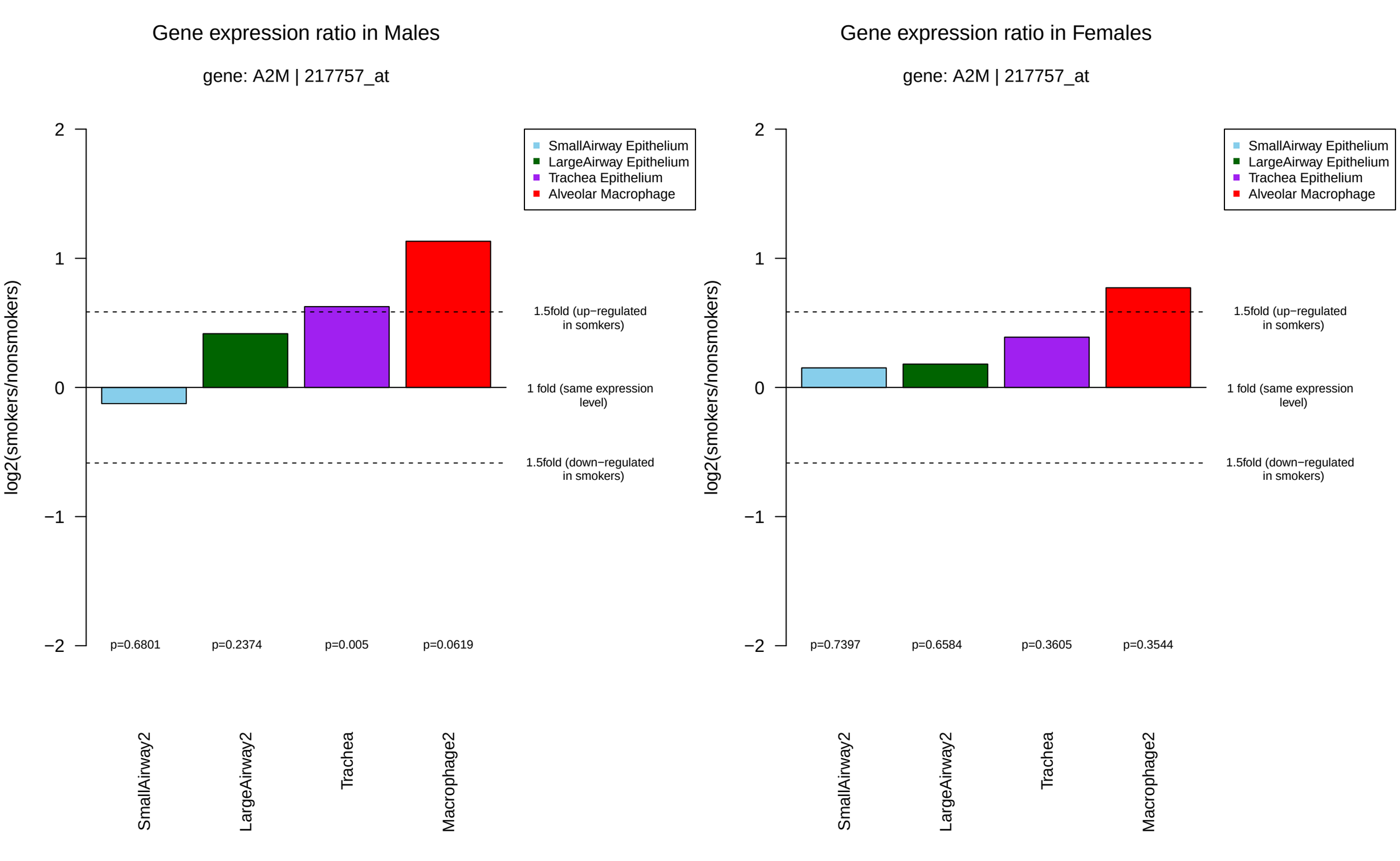
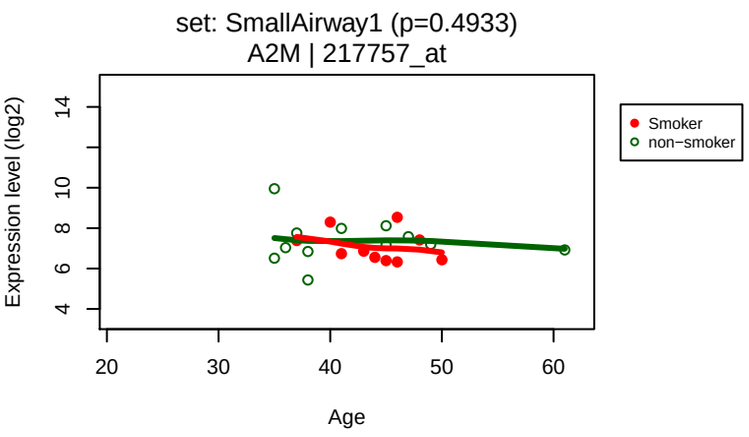
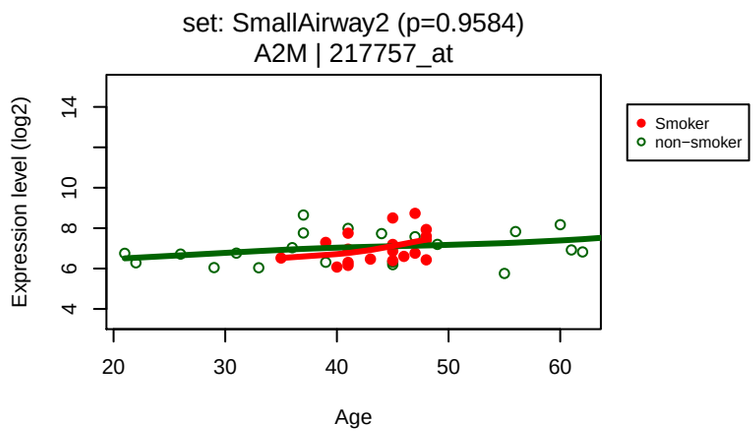
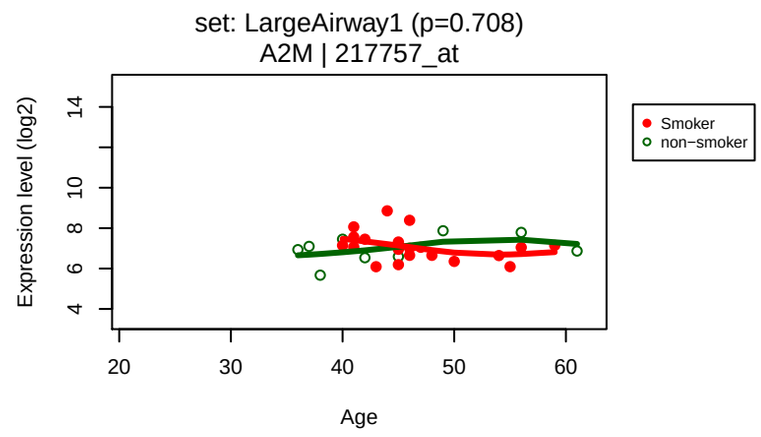

Gene expression of smokers and nonsmokers
gene: A2M | 217757_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4933
p=0.9584
p=0.2974
p=0.708
p=0.1459
p=0.0012
p=0.1852
p=0.031
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

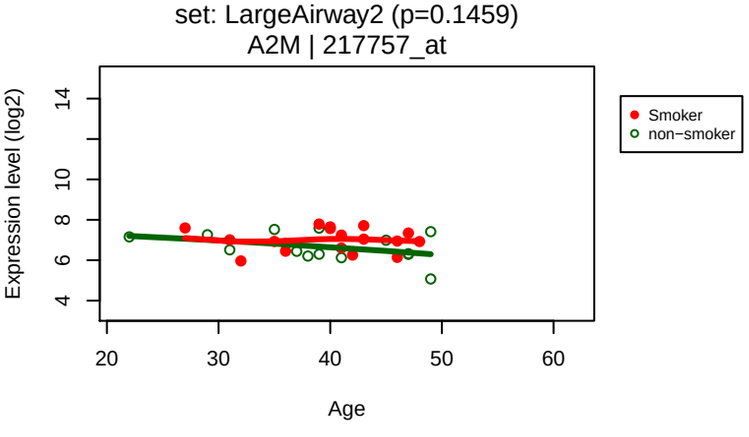

Gene expression ratio (smokers/nonsmokers)
gene: A2M | 217757_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4933
p=0.9584
p=0.2974
p=0.708
p=0.1459
p=0.0012
p=0.1852
p=0.031
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

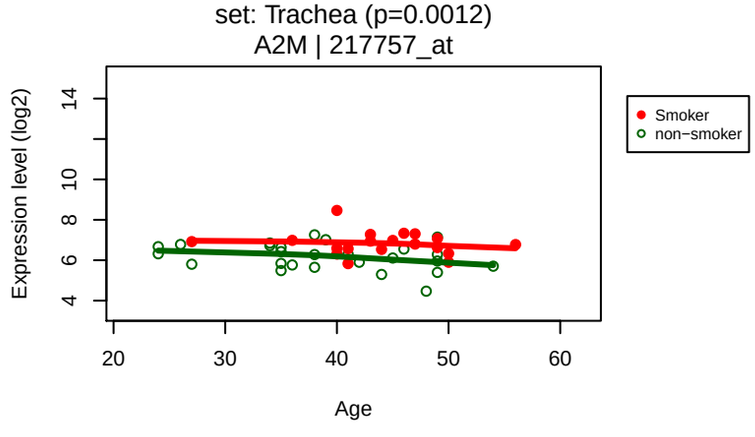

Gene expression ratio in Males
gene: A2M | 217757_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6801
p=0.2374
p=0.005
p=0.0619
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: A2M | 217757_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7397
p=0.6584
p=0.3605
p=0.3544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

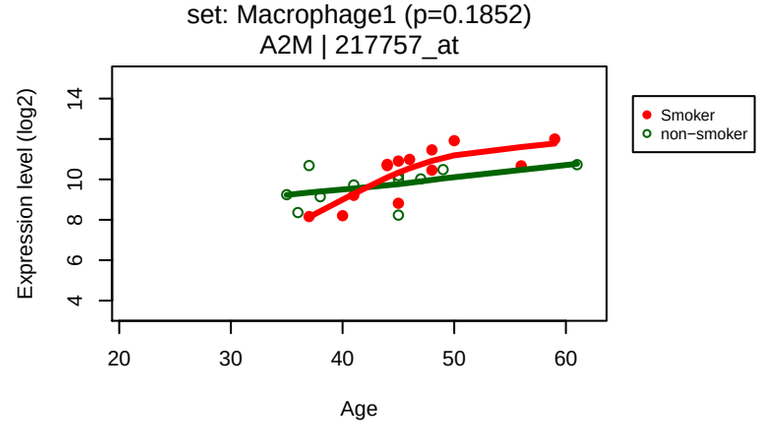

set: SmallAirway1 (p=0.4933)
A2M | 217757_at
Expression level (log2)
Age
Smoker
non−smoker

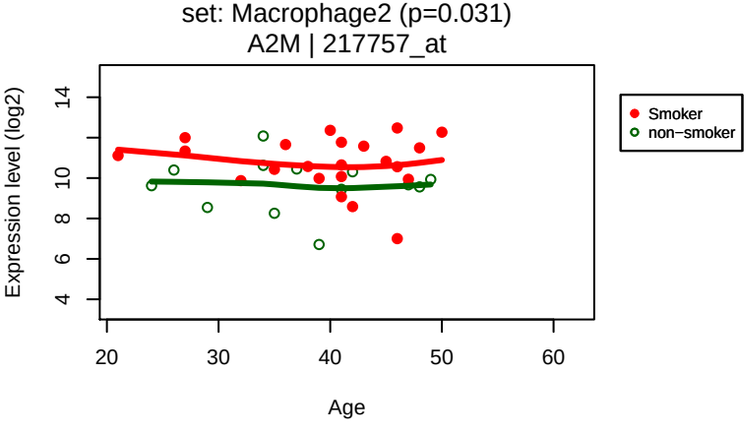

set: SmallAirway2 (p=0.9584)
A2M | 217757_at
Expression level (log2)
Age
Smoker
non−smoker

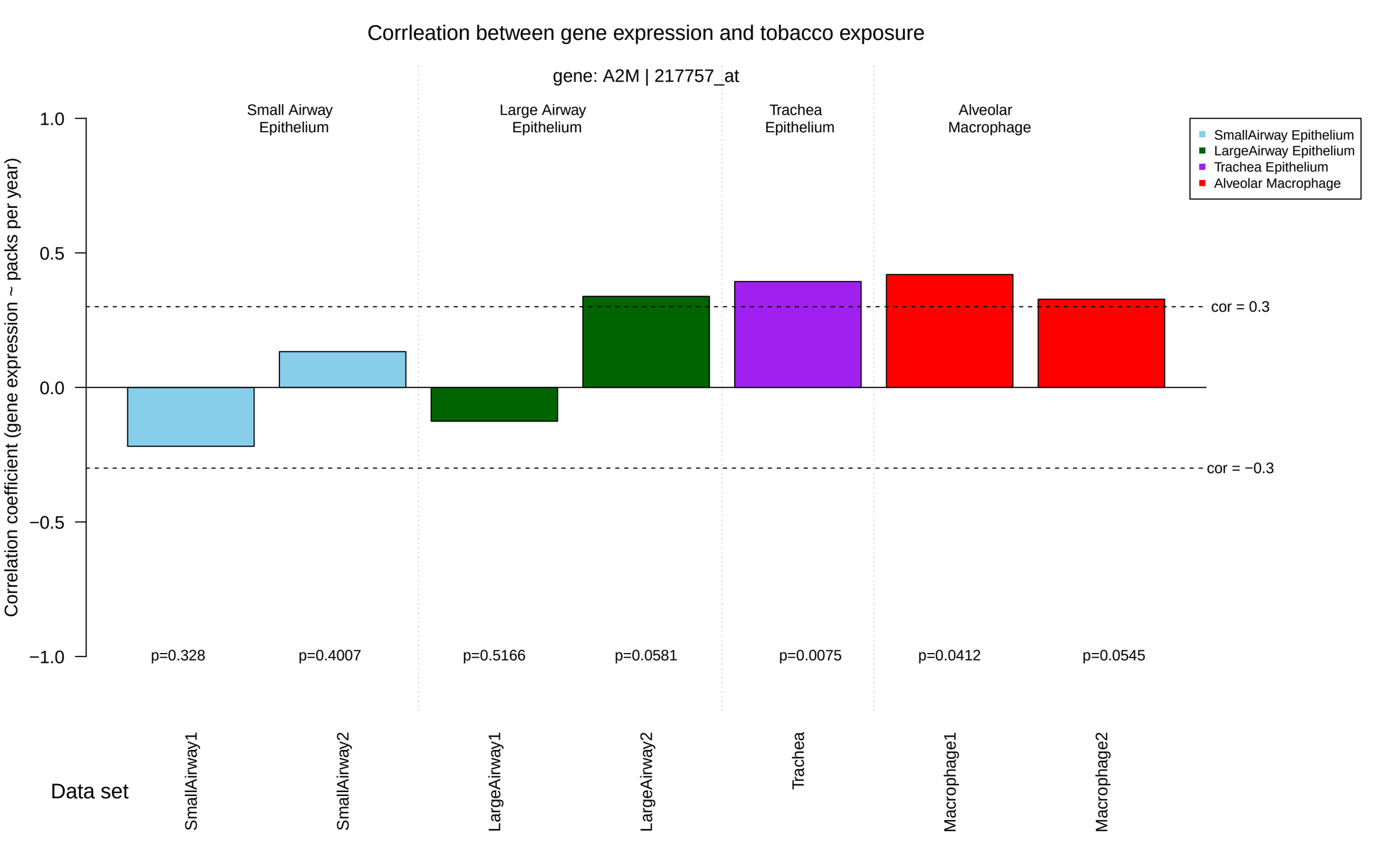

set: LargeAirway1 (p=0.708)
A2M | 217757_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1459)
A2M | 217757_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
A2M | 217757_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1852)
A2M | 217757_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.031)
A2M | 217757_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: A2M | 217757_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.328
p=0.4007
p=0.5166
p=0.0581
p=0.0075
p=0.0412
p=0.0545
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2